

Spherical Freezing Analogy for Cosmology

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Abstract

We develop a spherical freezing analogy for cosmological expansion. In the isotropic limit, a Stefan-type growth law for the ice radius gives $R(t) \propto t^{1/2}$ and hence $H_R^2 \propto R^{-4}$, matching the scaling of a radiation-dominated Friedmann universe. We then introduce an actively cooled isothermal core, which allows freezing in a warmer water bath and provides a more controllable experimental setting. Weak anisotropy is described through a quadrupolar deformation of the interface and related to anisotropic expansion variables. Finally, phase-field AFiD-MuRPhFi simulations validate the isotropic scaling and show that strong-core freshwater convection leaves the interface nearly circular, while controlled wall-forced conduction produces a robust leading elliptic deformation.

1 Introduction

Analogue models are useful when an evolution equation in one physical system has the same structure as an equation in another. The planar freezing-lake analogy [1, 2] shows that Stefan growth can reproduce cosmological expansion laws. Here we ask whether a fully spherical freezing geometry can realize the radiation-era Friedmann scaling, and whether controlled anisotropies of the freezing front can be used to mimic anisotropic expansion. The project is intentionally minimal: we first derive reduced Stefan models, then test the key scaling and deformation mechanisms with phase-field simulations. The simulations are not meant to be a complete experimental design; rather, they provide a numerical check of which terms in the reduced model can be made visible in a realistic heat-transfer setting.

2 Spherical freezing model

The planar freezing-lake model provides an exact analogue of a radiation-dominated Friedmann equation through the evolution of the ice thickness $s(t)$. [3] Here we show that an analogous result arises in a fully three-dimensional geometry by considering a spherically symmetric freezing problem.

Setup. Consider an ice sphere of radius $R(t)$ growing into an infinite bath of supercooled water. The liquid temperature far from the interface is

$$T(r \rightarrow \infty) = T_\infty < T_\phi,$$

where T_ϕ is the equilibrium freezing temperature. The ice-water interface is located at

$$r = R(t),$$

with

$$T(R(t), t) = T_\phi.$$

We assume:

1. spherical symmetry,
2. quasi-static heat conduction,
3. negligible temperature gradient inside the ice.

Temperature field in the liquid. In the freezing problem the temperature field is governed by the heat diffusion equation

$$\frac{\partial T}{\partial t} = \kappa \nabla^2 T,$$

where κ is the thermal diffusivity. In typical solidification processes the thermal diffusion time scale is much shorter than the time scale over which the interface position $R(t)$ evolves. Consequently the temperature field relaxes rapidly to a quasi-steady profile compared with the slow motion of the freezing front. Under this quasi-static approximation the time derivative may be neglected, reducing the heat equation to Laplace's equation

$$\nabla^2 T = 0,$$

which determines the instantaneous temperature distribution in each phase. Considering spherical symmetry, in the liquid region $r > R(t)$ the temperature obeys Laplace's equation

$$\nabla^2 T = \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dT}{dr} \right) = 0.$$

The spherically symmetric solution is

$$T_w(r) = A + \frac{B}{r}.$$

Applying the boundary conditions

$$T_w(R) = T_\phi, \quad T_w(\infty) = T_\infty,$$

gives

$$T_w(r) = T_\infty + (T_\phi - T_\infty) \frac{R}{r}.$$

The radial gradient is therefore

$$\frac{dT_w}{dr} = -(T_\phi - T_\infty) \frac{R}{r^2}.$$

At the interface

$$\left. \frac{dT_w}{dr} \right|_{R^+} = -\frac{T_\phi - T_\infty}{R}.$$

Stefan condition. The latent heat released during solidification must be conducted away through the liquid. The Stefan condition therefore gives

$$\rho_i L \dot{R} = -\lambda_w \left. \frac{dT_w}{dr} \right|_{R^+},$$

where ρ_i is the ice density, L the latent heat of fusion, and λ_w the thermal conductivity of water.

Substituting the interface gradient yields

$$\rho_i L \dot{R} = \lambda_w \frac{T_\phi - T_\infty}{R}.$$

Defining the constant

$$\alpha \equiv \frac{\lambda_w (T_\phi - T_\infty)}{\rho_i L},$$

we obtain the spherical Stefan growth law

$$\dot{R} = \frac{\alpha}{R}. \tag{1}$$

Exact solution. Multiplying Eq. (1) by R gives

$$R\dot{R} = \alpha.$$

Hence

$$\frac{d}{dt}(R^2) = 2\alpha,$$

so that

$$R^2(t) = R_0^2 + 2\alpha(t - t_0). \quad (2)$$

At late times the growth law becomes

$$R(t) \propto t^{1/2}.$$

Friedmann analogue. Define the spherical Hubble parameter

$$H_R \equiv \frac{\dot{R}}{R}.$$

Using Eq. (1) we obtain

$$H_R = \frac{\alpha}{R^2}, \quad H_R^2 = \alpha^2 R^{-4}. \quad (3)$$

This has exactly the form of a Friedmann equation for a spatially flat universe containing radiation,

$$H^2 \propto a^{-4}.$$

With the identification

$$a(t) \longleftrightarrow R(t),$$

the spherical freezing problem therefore reproduces the radiation-era expansion law

$$a(t) \propto t^{1/2}.$$

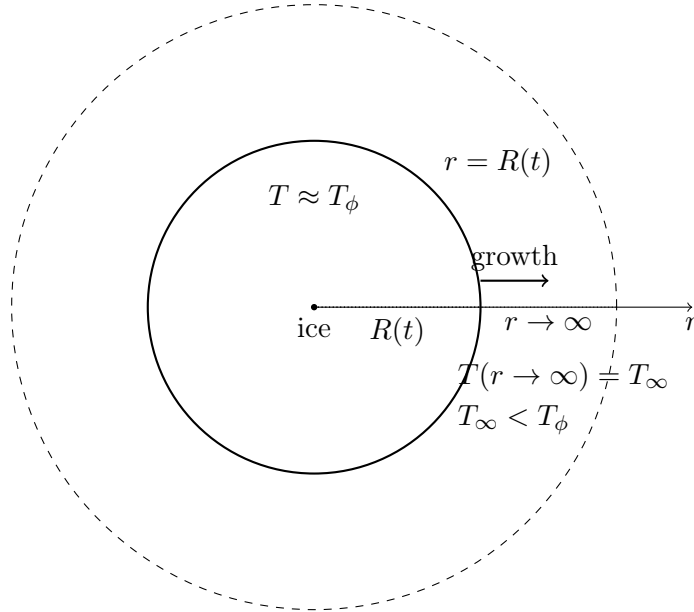


Figure 1: Spherical freezing model

Effective energy density and equation of state. Writing the evolution equation as

$$H_R^2 = \frac{8\pi G}{3} \rho_{\text{eff}}(R),$$

one finds

$$\rho_{\text{eff}}(R) = \frac{3\alpha^2}{8\pi G} R^{-4}.$$

Comparing with

$$\rho(a) \propto a^{-3(1+w)},$$

gives $w = \frac{1}{3}$.

Thus the spherical freezing model with an infinite thermal reservoir provides an *exact* analogue of a radiation-dominated Friedmann universe. This result parallels the planar freezing-lake model, but arises from the geometric spreading of radial heat flux rather than from the growth of a planar conductive layer.

3 Spherical freezing with an actively cooled isothermal core

The previous spherical model assumes an ice sphere growing into supercooled water, $T_\infty < T_\phi$, so that the latent heat released at the interface is conducted away through the liquid. This gives a one-sided Stefan condition and leads directly to the radiation-like scaling $H_R^2 \propto R^{-4}$.

A more experimentally controllable realization is to place a small actively cooled core at the center of the growing ice sphere. The core is maintained at a fixed temperature

$$T_0 < T_\phi$$

by an external cooling mechanism. The surrounding water bath may now be warmer than the freezing point,

$$T_\infty > T_\phi,$$

so that the liquid tends to melt the interface, while the cooled core removes heat and drives freezing.

Setup. Consider a small spherical core of radius a , with

$$a \ll R(t),$$

placed at the center of an ice shell of outer radius $R(t)$. The core is assumed to be thermostatted at a fixed temperature $T_0 < T_\phi$. Physically this may be realized by a high-conductivity metallic sphere or a small heat exchanger connected to an external cooling bath. The ice-water interface remains at

$$T(R(t), t) = T_\phi.$$

We assume:

1. spherical symmetry,
2. quasi-static heat transport,
3. an effectively isothermal cooled core at temperature T_0 ,
4. a reduced thermal-resistance description of the ice-side heat flux.

Effective ice-side heat flux. Instead of resolving the detailed heat-transfer mechanism inside the cooled core, we model the core and its cooling apparatus as an effective heat sink. Thus we write

$$q_i^{\text{eff}} = \lambda_i \frac{T_\phi - T_0}{R}, \quad (4)$$

where λ_i is the thermal conductivity of ice.

Water-side heat flux. In the water region, assuming spherical symmetry and quasi-static heat conduction, the temperature field satisfies

$$\nabla^2 T_w = 0,$$

with boundary conditions

$$T_w(R) = T_\phi, \quad T_w(\infty) = T_\infty.$$

Hence

$$T_w(r) = T_\infty + (T_\phi - T_\infty) \frac{R}{r}.$$

The radial derivative at the interface is

$$\left. \frac{dT_w}{dr} \right|_{R^+} = -\frac{T_\phi - T_\infty}{R}.$$

Therefore the water-side contribution to the Stefan balance is

$$q_w = -\lambda_w \left. \frac{dT_w}{dr} \right|_{R^+} = \lambda_w \frac{T_\phi - T_\infty}{R}. \quad (5)$$

If the bath is warmer than the freezing point, $T_\infty > T_\phi$, then this term is negative and opposes freezing.

Two-sided effective Stefan balance. The latent heat released at the moving interface is balanced by the net heat flux away from the interface:

$$\rho_i L \dot{R} = q_i^{\text{eff}} + q_w. \quad (6)$$

Using Eqs. (4) and (5), we obtain

$$\rho_i L \dot{R} = \lambda_i \frac{T_\phi - T_0}{R} + \lambda_w \frac{T_\phi - T_\infty}{R}. \quad (7)$$

Define

$$\alpha_{\text{core}} \equiv \frac{\lambda_i(T_\phi - T_0) + \lambda_w(T_\phi - T_\infty)}{\rho_i L}. \quad (8)$$

Then Eq. (7) becomes

$$\dot{R} = \frac{\alpha_{\text{core}}}{R}. \quad (9)$$

Growth condition. Freezing requires

$$\alpha_{\text{core}} > 0.$$

If $T_\infty > T_\phi$, this means

$$\lambda_i(T_\phi - T_0) > \lambda_w(T_\infty - T_\phi). \quad (10)$$

Thus the cooled core must remove heat faster than the warm water bath supplies it to the interface.

Friedmann analogue. Equation (9) has the same mathematical form as the supercooled-bath Stefan law. Therefore

$$R\dot{R} = \alpha_{\text{core}},$$

and

$$R^2(t) = R_0^2 + 2\alpha_{\text{core}}(t - t_0). \quad (11)$$

The spherical Hubble parameter is

$$H_R = \frac{\dot{R}}{R} = \frac{\alpha_{\text{core}}}{R^2},$$

so that

$$H_R^2 = \alpha_{\text{core}}^2 R^{-4}. \quad (12)$$

Thus the actively cooled-core model also realizes the radiation-like Friedmann scaling

$$H^2 \propto a^{-4},$$

with the identification

$$a(t) \longleftrightarrow R(t).$$

Experimental realization. A possible laboratory setup consists of a small spherical metallic core or heat exchanger placed at the center of a transparent water tank. The core is connected to an external cooling loop which maintains it at a fixed temperature $T_0 < T_\phi$. A thin initial ice layer is first formed around the core, and the subsequent growth of the ice shell is monitored optically.

The water bath temperature T_∞ can be chosen above T_ϕ to study freezing driven primarily by heat extraction through the cooled core, testing the condition

$$\lambda_i(T_\phi - T_0) > \lambda_w(T_\infty - T_\phi).$$

The main observables are:

$$R(t), \quad \dot{R}(t), \quad H_R(t) = \frac{\dot{R}}{R}.$$

The key scaling test is

$$R^2(t) \simeq R_0^2 + 2\alpha_{\text{core}}t, \quad (13)$$

or equivalently

$$H_R^2 R^4 \simeq \alpha_{\text{core}}^2 = \text{constant}. \quad (14)$$

Numerical simulation The evolution of the ice radius in the model is compared with that of the scale factor in a simple CDM universe in Fig 3, constants are provided in Table 1

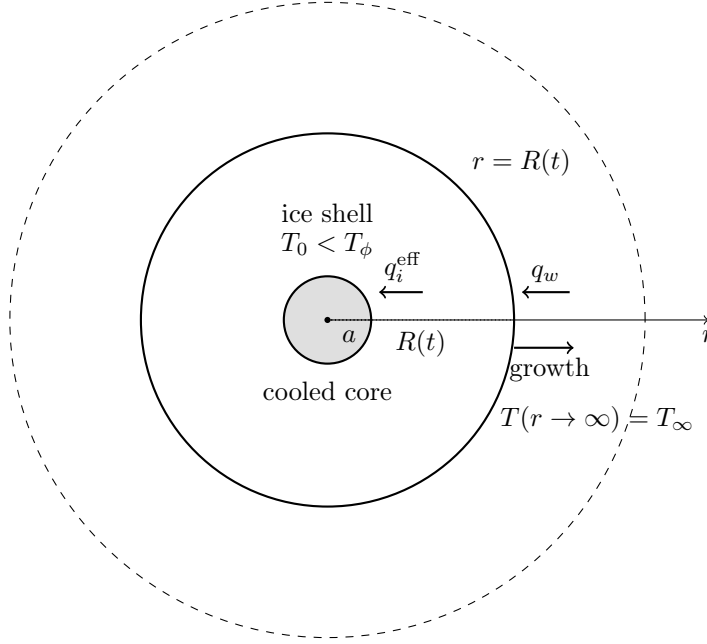


Figure 2: Actively cooled isothermal-core model.

Symbol	Value	Unit	Description / Source
ρ_i	917	kg m^{-3}	Density of ice near 0°C .
L	3.34×10^5	J kg^{-1}	Latent heat of fusion of ice
λ_i	2.22	$\text{W m}^{-1} \text{K}^{-1}$	Thermal conductivity of ice
λ_w	0.56	$\text{W m}^{-1} \text{K}^{-1}$	Thermal conductivity of water
T_ϕ	0.0	$^\circ\text{C}$	Ice–water interface temperature.
T_∞	3.0	$^\circ\text{C}$	Bulk water-bath temperature.
T_0	−20.0	$^\circ\text{C}$	Maintained temperature of the cooled core.
a	2.0×10^{-3}	m	Radius of the actively cooled core.
h_w	5.0	$\text{W m}^{-2} \text{K}^{-1}$	water-side convective heat-transfer coefficient.

Table 1: Physical constants and model parameters used in the numerical simulations of the actively cooled spherical freezing model.

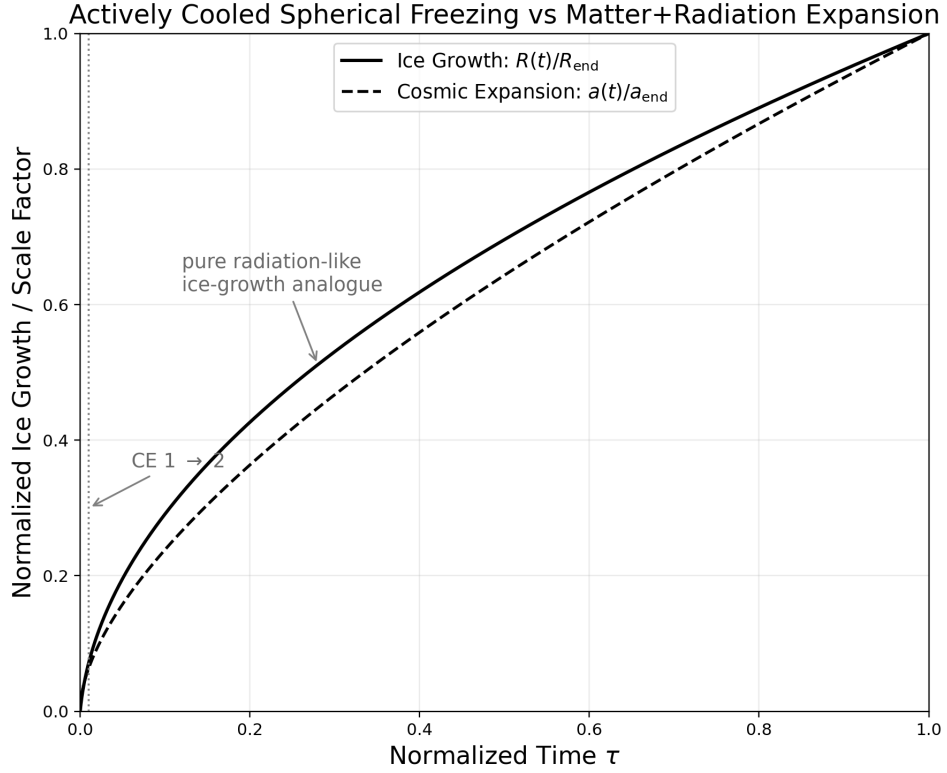


Figure 3: Normalized ice thickness $R(t)/R_{\text{end}}$ in the spherical freezing with an actively cooled isothermal core (solid curve) compared with the normalized scale factor $a(t)/a_0$ in a matter–radiation cosmological expansion (CE) model with zero cosmological constant ($\Lambda = 0$), both plotted as functions of normalized time τ .

4 Actively cooled spherical freezing with buoyancy-driven convection

We now combine the actively cooled-core model with buoyancy-driven heat transport in the surrounding liquid. The cooled core removes heat through the ice shell, while the water-side flux is enhanced and distorted by convection. Gravity selects a preferred vertical axis, so the interface is naturally described as a weakly deformed axisymmetric sphere.

Axisymmetric interface. We parametrize the ice–water interface by

$$R(\theta, t) = S(t) [1 + \varepsilon(t) P_2(\cos \theta)], \quad |\varepsilon| \ll 1, \quad (15)$$

where $S(t)$ is the mean radius, $\varepsilon(t)$ is the quadrupolar anisotropy, and

$$P_2(\mu) = \frac{1}{2}(3\mu^2 - 1). \quad (16)$$

The polar and equatorial radii are therefore, to linear order,

$$R_{\parallel} = S(1 + \varepsilon), \quad R_{\perp} = S \left(1 - \frac{\varepsilon}{2}\right). \quad (17)$$

This is the same kinematical structure as a locally rotationally symmetric Bianchi-I geometry, where one direction expands differently from the two transverse directions [4].

Effective Stefan condition. The local Stefan balance is taken to be

$$\rho_i L \partial_t R(\theta, t) = \lambda_i \frac{T_{\phi} - T_0}{R(\theta, t)} + \lambda_w \frac{T_{\phi} - T_{\infty}}{R(\theta, t)} Nu(\theta, t), \quad (18)$$

where $T_0 < T_{\phi}$ is the actively maintained core temperature and $Nu(\theta, t)$ is the local water-side Nusselt factor. The first term describes effective heat extraction by the cooled core; the second describes heat exchange with the liquid. If $T_{\infty} > T_{\phi}$, then the water-side term is negative and opposes freezing.

Define

$$A_i = \frac{\lambda_i(T_{\phi} - T_0)}{\rho_i L}, \quad A_w = \frac{\lambda_w(T_{\phi} - T_{\infty})}{\rho_i L}. \quad (19)$$

Then

$$\partial_t R(\theta, t) = \frac{A_i + A_w Nu(\theta, t)}{R(\theta, t)}. \quad (20)$$

We model the convective anisotropy by

$$Nu(\theta, t) = \bar{Nu}(S) [1 + \xi(S) P_2(\cos \theta)]. \quad (21)$$

This reduced Nusselt description is not meant to resolve the full hydrodynamic flow; it encodes the net effect of water-side transport on the moving interface. Such Nusselt–Rayleigh closures are standard in buoyancy-driven heat transfer, where one often writes $Nu \sim Ra^{\beta}$ over a given scaling regime [5]. The anisotropic part, controlled by $\xi(S)$, plays the role of a small effective anisotropic stress source, analogous to the skew-stress sources studied in anisotropic cosmology [6, 7, 8].

Introducing

$$\Lambda(S) = A_i + A_w \bar{Nu}(S), \quad \mathcal{X}(S) = A_w \bar{Nu}(S) \xi(S), \quad (22)$$

Eq. (20) becomes

$$\partial_t R(\theta, t) = \frac{\Lambda(S) + \mathcal{X}(S) P_2(\cos \theta)}{R(\theta, t)}. \quad (23)$$

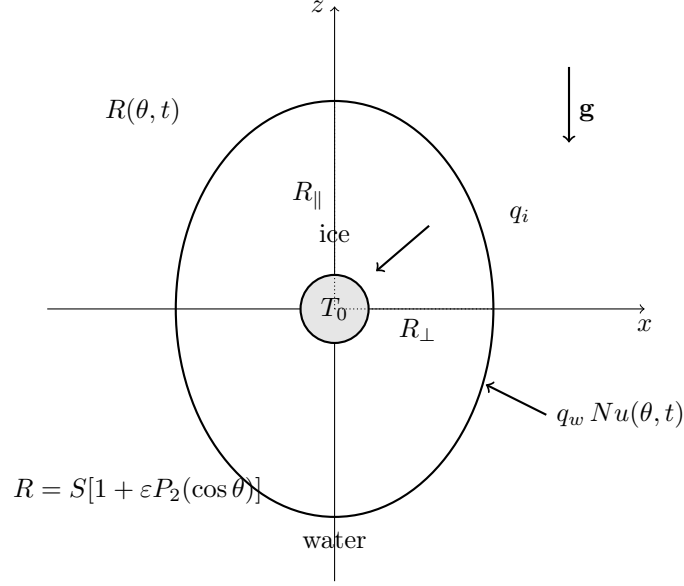


Figure 4: Actively cooled spherical freezing with anisotropic water-side convection.

Reduced evolution equations. Substituting Eq. (15) into Eq. (23) and expanding to first order in ε gives

$$\partial_t R = \dot{S} + \left(\dot{S} \varepsilon + S \dot{\varepsilon} \right) P_2, \quad (24)$$

while

$$\frac{\Lambda + \mathcal{X} P_2}{R} \simeq \frac{\Lambda}{S} + \frac{\mathcal{X} - \Lambda \varepsilon}{S} P_2. \quad (25)$$

Equating the isotropic and quadrupolar parts yields

$$\dot{S} = \frac{\Lambda(S)}{S}, \quad (26)$$

and

$$\dot{\varepsilon} = \frac{1}{S^2} [\mathcal{X}(S) - 2\Lambda(S)\varepsilon]. \quad (27)$$

Equivalently,

$$\dot{\varepsilon} = \frac{\Lambda(S)}{S^2} [\xi_{\text{eff}}(S) - 2\varepsilon], \quad (28)$$

where

$$\xi_{\text{eff}}(S) = \frac{\mathcal{X}(S)}{\Lambda(S)} = \frac{A_w \bar{N} u(S)}{A_i + A_w \bar{N} u(S)} \xi(S). \quad (29)$$

Thus the isotropic cooled-core flux screens the anisotropic forcing generated on the water side.

Mean Friedmann analogue. Assume the reduced convective scaling

$$Ra \propto S^3, \quad \bar{N} u(S) = Nu_0 S^{3\beta}, \quad B \equiv A_w Nu_0. \quad (30)$$

Then

$$\Lambda(S) = A_i + B S^{3\beta}, \quad (31)$$

and

$$H \equiv \frac{\dot{S}}{S} = \left(A_i + B S^{3\beta} \right) S^{-2}. \quad (32)$$

Therefore

$$H^2 = A_i^2 S^{-4} + 2A_i B S^{3\beta-4} + B^2 S^{6\beta-4}. \quad (33)$$

The cooled core gives the radiation-like term S^{-4} , while water-side convection generates the additional mixed and convective contributions. For $\beta = 1/3$,

$$H^2 = A_i^2 S^{-4} + 2A_i B S^{-3} + B^2 S^{-2}, \quad (34)$$

which has the formal structure of radiation-, matter-, and curvature-like terms. This identification should be understood as a structural analogy between evolution equations, not as a literal gravitational dynamics.

Anisotropic expansion and shear. Define directional scale factors

$$a_{\parallel} = S(1 + \varepsilon), \quad a_{\perp} = S \left(1 - \frac{\varepsilon}{2}\right), \quad (35)$$

so that $a_{\parallel} a_{\perp}^2 \simeq S^3$. Their Hubble rates are

$$H_{\parallel} \simeq H + \dot{\varepsilon}, \quad H_{\perp} \simeq H - \frac{1}{2}\dot{\varepsilon}. \quad (36)$$

The LRS shear scalar is

$$\sigma^2 = \frac{1}{3}(H_{\parallel} - H_{\perp})^2 = \frac{3}{4}\dot{\varepsilon}^2. \quad (37)$$

This is the same shear construction used in LRS Bianchi-I cosmology [4].

If the anisotropic forcing decays, $\xi_{\text{eff}} \rightarrow 0$, then Eq. (27) gives

$$\frac{d\varepsilon}{dS} = -\frac{2\varepsilon}{S}, \quad (38)$$

and hence

$$\varepsilon = \varepsilon_0 S^{-2}. \quad (39)$$

Using Eq. (26),

$$\dot{\varepsilon} = -2\varepsilon_0 \left(A_i + BS^{3\beta}\right) S^{-4}, \quad (40)$$

so

$$\sigma^2 = 3\varepsilon_0^2 \left(A_i + BS^{3\beta}\right)^2 S^{-8}. \quad (41)$$

The conservative anisotropic Friedmann form is therefore

$$H^2 = \left(A_i + BS^{3\beta}\right)^2 S^{-4} + \varepsilon_0^2 \left(A_i + BS^{3\beta}\right)^2 S^{-8}. \quad (42)$$

The first term represents the mean expansion analogue, while the second term plays the role of the shear contribution. In the core-dominated regime one recovers

$$H^2 \propto S^{-4}, \quad \sigma^2 \propto S^{-8}.$$

In the convection-dominated regime,

$$H^2 \propto S^{6\beta-4}, \quad \sigma^2 \propto S^{6\beta-8}.$$

For $\beta = 1/3$, the shear term scales as S^{-6} , matching the standard LRS Bianchi-I shear scaling [4]. More generally, when ξ_{eff} remains nonzero, the model is better interpreted as a sourced anisotropic expansion, analogous to Bianchi-I cosmologies with anisotropic stresses rather than a freely decaying shear sector [6, 7, 8].

5 Pure-conduction anisotropy in a square box

The convection model is not the only way to generate anisotropic freezing. A controlled anisotropy can also be produced by pure conduction if the outer thermal boundary is anisotropic. This is closer to the wall-forced simulations below, where the x -walls are kept warmer than the transverse direction.

Here we give a simple two-dimensional analytic model. The exact problem of a circular freezing interface inside a square box does not have a simple elementary closed-form solution. However, the leading anisotropic term can be obtained explicitly by solving the square-box conduction problem and expanding the temperature field near the centre.

Square-box conduction field. Consider a square box

$$-a < x < a, \quad -a < y < a,$$

with fixed boundary temperatures

$$T(\pm a, y) = T_x, \quad T(x, \pm a) = T_y.$$

The x -walls are warmer when

$$T_x > T_y.$$

Let

$$\Delta T = T_x - T_y.$$

We write

$$T(x, y) = T_y + \Delta T u(x, y),$$

where u satisfies

$$\nabla^2 u = 0, \quad u(\pm a, y) = 1, \quad u(x, \pm a) = 0.$$

A separated solution is

$$u(x, y) = \sum_{n=0}^{\infty} \frac{4(-1)^n}{(2n+1)\pi} \frac{\cosh(k_n x)}{\cosh(k_n a)} \cos(k_n y),$$

where

$$k_n = \frac{(2n+1)\pi}{2a}.$$

This is an exact Fourier-series solution for the square-box conduction problem.

Near the centre of the box, symmetry implies that the first anisotropic term is quadrupolar:

$$u(x, y) = \frac{1}{2} + \Gamma_2(x^2 - y^2) + O(r^4),$$

where

$$\Gamma_2 = \frac{\pi}{2a^2} \sum_{n=0}^{\infty} \frac{(-1)^n(2n+1)}{\cosh[(2n+1)\pi/2]}.$$

Numerically,

$$\Gamma_2 \simeq \frac{0.548}{a^2}.$$

Using polar coordinates centred on the ice,

$$x = r \cos \theta, \quad y = r \sin \theta,$$

we have

$$x^2 - y^2 = r^2 \cos 2\theta.$$

Thus, near the interface, the imposed square-box temperature field contains the quadrupolar contribution

$$T_{\text{quad}}(r, \theta) = \Delta T \Gamma_2 r^2 \cos 2\theta.$$

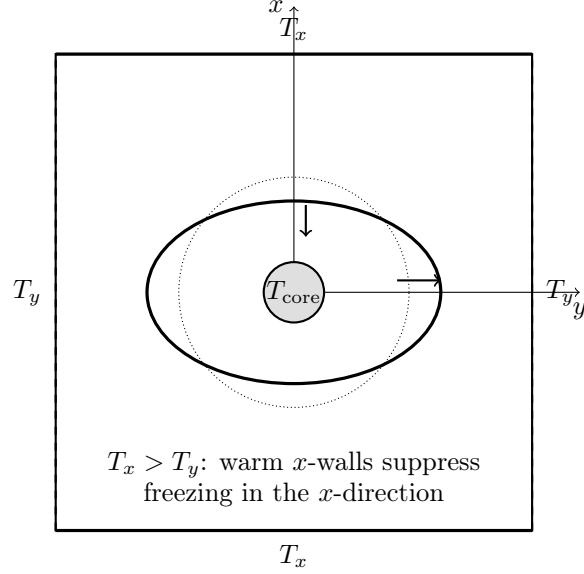


Figure 5: Two-dimensional pure-conduction anisotropy in a square box. Warm x -walls produce a quadrupolar temperature field near the centre. The stronger water-side heat input in the x -direction suppresses freezing there, leading to an x -flattened and y -elongated interface.

Circular interface response. Now place a circular ice–water interface at

$$r = R(t),$$

with

$$T(R, \theta) = T_\phi.$$

The quadrupolar part of the water temperature must vanish at the interface. The harmonic continuation of the square-box quadrupole is therefore

$$T_2(r, \theta) = \Delta T \Gamma_2 \left(r^2 - \frac{R^4}{r^2} \right) \cos 2\theta.$$

Indeed,

$$T_2(R, \theta) = 0,$$

and for r outside the interface it matches the imposed local quadrupolar field. The corresponding radial gradient at the interface is

$$\left. \frac{\partial T_2}{\partial r} \right|_{R^+} = 4\Delta T \Gamma_2 R \cos 2\theta.$$

The local Stefan condition can then be written as

$$\rho_i L \partial_t R(\theta, t) = q_{\text{core}}(R) - \lambda_w \left. \frac{\partial T_w}{\partial r} \right|_{R^+}.$$

Keeping only the anisotropic part gives

$$\partial_t R(\theta, t) = V_0(R) - \frac{4\lambda_w \Delta T \Gamma_2 R}{\rho_i L} \cos 2\theta.$$

Equivalently,

$$\partial_t R(\theta, t) = V_0(R) + V_2(R) \cos 2\theta,$$

with

$$V_2(R) = -\frac{4\lambda_w \Delta T \Gamma_2 R}{\rho_i L}.$$

For warm x -walls,

$$T_x > T_y, \quad \Delta T > 0,$$

so

$$V_2(R) < 0.$$

Along the x -axis,

$$\theta = 0, \pi, \quad \cos 2\theta = 1,$$

and therefore

$$\partial_t R_x = V_0 + V_2 < V_0.$$

Along the y -axis,

$$\theta = \frac{\pi}{2}, \frac{3\pi}{2}, \quad \cos 2\theta = -1,$$

so

$$\partial_t R_y = V_0 - V_2 > V_0.$$

Thus the analytic model predicts

$$R_x < R_y,$$

namely an interface that is flattened in the x -direction and elongated in the y -direction.

Weak-deformation equation. Writing

$$R(\theta, t) = S(t) [1 + \varepsilon(t) \cos 2\theta], \quad |\varepsilon| \ll 1,$$

and expanding to first order gives

$$\dot{S} = V_0(S),$$

and

$$\dot{\varepsilon} = \frac{V_2(S)}{S} + \varepsilon \left[V_0'(S) - \frac{V_0(S)}{S} \right].$$

The first term is the imposed conductive quadrupole source. The second term contains both the geometric dilution of a relative deformation and the radial dependence of the mean growth law.

This expression is general and does not assume a particular form for $V_0(S)$. If the mean dynamics are

$$V_0(S) = \dot{S} = \frac{\alpha}{S},$$

or equivalently

$$S^2(t) \propto t,$$

then

$$V_0'(S) = -\frac{V_0(S)}{S}.$$

The deformation equation becomes

$$\dot{\varepsilon} = \frac{V_2(S)}{S} - 2\frac{\dot{S}}{S}\varepsilon.$$

Defining the Hubble-like rate

$$H \equiv \frac{\dot{S}}{S},$$

and the dimensionless conductive anisotropy source

$$\xi_{\text{cond}}(S) \equiv \frac{V_2(S)}{V_0(S)},$$

we may write

$$\dot{\varepsilon} = H [\xi_{\text{cond}}(S) - 2\varepsilon].$$

This has the same structure as the anisotropy equation in the convection model: a quadrupolar source term drives the deformation, while the term $-2H\varepsilon$ dilutes it as the mean radius grows. The difference is only the physical origin of the source: here ξ_{cond} is imposed by a conductive boundary anisotropy, whereas in the convection model the analogous source comes from anisotropic water-side heat transport.

6 AFiD Simulation

We validate the actively cooled-core freezing model using two-dimensional phase-field simulations performed with AFiD-MuRPhFi[9]. The simulation study has two purposes. First, we test the actively cooled-core model in the isotropic no-flow limit and verify the Stefan-type scaling

$$\dot{S} \propto \frac{1}{S}, \quad H^2 \propto S^{-4}.$$

Second, we add freshwater buoyancy-driven convection and test whether the resulting natural flow modifies the mean freezing rate or visibly deforms the ice–water interface.

Phase-field representation. The solid–liquid interface is represented by a phase-field variable ϕ , with

$$\phi = 1 \quad \text{ice}, \quad \phi = 0 \quad \text{water}.$$

The ice–water interface is extracted from the contour

$$\phi = \frac{1}{2}.$$

The mean radius of the ice shell is denoted by $S(t)$ and is obtained from the extracted interface.

Actively cooled-core setup. A circular actively cooled core is placed at the centre of the domain. The core temperature, freezing temperature, and initial water temperature are

$$T_{\text{core}} = -20^\circ\text{C}, \quad T_\phi = 0^\circ\text{C}, \quad T_\infty = 3^\circ\text{C}.$$

The code uses the nondimensional temperature

$$\Theta = \frac{T - T_{\text{core}}}{T_\infty - T_{\text{core}}},$$

so that

$$\Theta_{\text{core}} = 0, \quad \Theta_\infty = 1, \quad \Theta_\phi = \frac{20}{23}.$$

The simulations are performed in a thin two-dimensional computational cell. In the convection simulations, the AFiD wall-normal coordinate x is used as the physical vertical direction. Thus gravity points downward along $-x$, while upward motion corresponds to the $+x$ direction.

Quantity	Value	Meaning
Domain	$1 \times 1 \times 0.01$	Thin 2D computational cell
Core radius	$r_{\text{core}} = 0.08$	Actively cooled core radius
Initial ice radius	$r_{\text{seed}} = 0.12$	Initial phase-field interface radius
Core temperature	$T_{\text{core}} = -20^\circ\text{C}$	Fixed isothermal core
Interface temperature	$T_\phi = 0^\circ\text{C}$	Freezing temperature
Water temperature	$T_\infty = 3^\circ\text{C}$	Initial surrounding water temperature
Phase field	$\phi = 1$ ice, $\phi = 0$ water	Interface at $\phi = 1/2$
Salinity	off	Freshwater case, no salt field
Initial ice temperature	conductive profile	Radial conductive shell profile

Table 2: Common active-core setup used in the AFiD-MuRPhFi simulations.

Two-dimensional mode diagnostics. The reduced analytic model above used the three-dimensional quadrupole coefficient ε in the expansion

$$R(\theta, t) = S(t) [1 + \varepsilon(t) P_2(\cos \theta)].$$

The AFiD calculations in this section are two-dimensional slices. We therefore diagnose the interface using a Fourier expansion of the extracted contour,

$$R(\theta, t) = S(t) \left[1 + \sum_{n \geq 1} (a_n(t) \cos n\theta + b_n(t) \sin n\theta) \right],$$

where θ is measured from the upward $+x$ direction in the simulation plane. The reported shape-mode amplitude is

$$M_n(t) = \sqrt{a_n^2(t) + b_n^2(t)}.$$

Thus M_2 is the two-dimensional analogue of a quadrupolar or elliptic deformation, while M_1 measures a dipolar/off-centre distortion. The mode M_4 is particularly important as a diagnostic because a circular interface represented on a Cartesian grid can produce a fourfold numerical leakage mode.

Similarly, the angular variation of the interfacial heat flux is written as a Fourier series. Let $q_{\text{int}}(\theta, t)$ denote the normal heat flux measured near the interface and let $\bar{q}_{\text{int}}(t)$ be its angular average. We define

$$\tilde{q}(\theta, t) = \frac{q_{\text{int}}(\theta, t)}{\bar{q}_{\text{int}}(t)} - 1,$$

and write

$$\tilde{q}(\theta, t) = \sum_{n \geq 1} [\xi_n(t) \cos n\theta + \eta_n(t) \sin n\theta].$$

The coefficient ξ_2 is the two-dimensional analogue of the quadrupolar Nusselt anisotropy ξ used in the reduced model. In the diagnostics below, the cosine coefficients ξ_n are reported because the vertical direction has been fixed by gravity. The sine components are monitored in the code but are not shown explicitly. Since the no-flow calculation contains small grid-induced modes, the convection results are compared to and, when appropriate, subtracted from the no-flow baseline at matched radius S .

No-flow Stefan validation. In the absence of buoyancy and fluid motion, the reduced actively cooled-core model predicts

$$\dot{S} = \frac{\alpha_{\text{core}}}{S}.$$

Therefore

$$S^2(t) = S_0^2 + 2\alpha_{\text{core}}t.$$

Equivalently, defining

$$H = \frac{\dot{S}}{S},$$

one expects

$$H^2 \propto S^{-4}, \quad H^2 S^4 \simeq \text{const.}$$

Thus the no-flow simulation provides a direct numerical test of the radiation-like Friedmann scaling of the actively cooled-core model.

We first run no-flow simulations at two phase-field resolutions, labelled 128 and 256. These labels refer to the refined scalar/phase-field grid resolution used for the circular interface extraction. In the multiresolution setup, the velocity/pressure grid is coarser than the refined scalar grid. The nominal Rayleigh and Prandtl numbers are still specified in the input files because AFiD uses nondimensional variables, but buoyancy is disabled by setting

$$active_T = 0.$$

Therefore the listed Ra should be understood as part of the nondimensional input convention rather than as an active buoyancy forcing in this validation run.

Quantity	Run 128	Run 256
Base grid	$64 \times 64 \times 1$	$128 \times 128 \times 1$
Refined grid	$128 \times 128 \times 1$	$256 \times 256 \times 1$
Nominal Rayleigh number	$Ra = 10^4$	$Ra = 10^4$
Prandtl number	$Pr = 1$	$Pr = 1$
Thermal buoyancy	off	off
Freshwater anomaly	off	off
Final time	$T_{\text{max}} = 0.08$	$T_{\text{max}} = 0.10$
Phase field	on	on
Salinity	off	off

Table 3: No-flow AFiD-MuRPhFi simulations used to validate the actively cooled-core Stefan scaling. Since $active_T = 0$, the listed Rayleigh number is nominal and does not drive buoyancy.

The linear fit of $S^2(t)$ gives

$$R_{\text{fit}}^2 \simeq 0.9999$$

for both tested resolutions, confirming that the phase-field simulation follows the expected Stefan-type growth law. A log-log fit of the post-transient data gives

$$H^2 \propto S^{-3.93}$$

for the lower-resolution run and

$$H^2 \propto S^{-4.01}$$

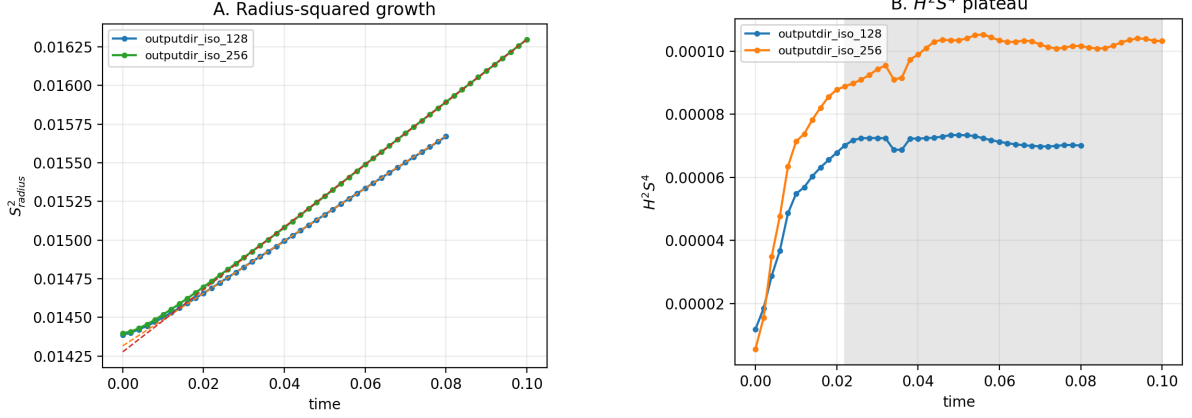
for the higher-resolution run. The second result is very close to the ideal prediction

$$H^2 \propto S^{-4}.$$

The corresponding fitted growth coefficients are

$$\alpha_{\text{fit}}^{128} \simeq 8.42 \times 10^{-3}, \quad \alpha_{\text{fit}}^{256} \simeq 9.98 \times 10^{-3},$$

in simulation units. Residual shape modes in the no-flow simulation are dominated by a fourfold contribution associated with the Cartesian grid and phase-field contour extraction. This mode is interpreted as a numerical baseline rather than a physical anisotropy.



(A) Radius-squared growth.

(B) Radiation-like invariant.

Figure 6: AFiD-MuRPhFi validation of the actively cooled-core freezing model in the no-flow limit. Panel (A) shows that $S^2(t)$ is approximately linear after the short phase-field relaxation transient, consistent with $\dot{S} \propto 1/S$. Panel (B) shows that $H^2 S^4$ approaches a post-transient plateau, confirming the radiation-like scaling $H^2 \propto S^{-4}$.

Freshwater buoyancy model. We then include buoyancy-driven convection in the surrounding water. Since the water temperature lies close to the density maximum of freshwater, a standard linear Boussinesq law is not appropriate. Instead, we use a reduced freshwater density-anomaly law

$$\rho(T) = \rho_0 [1 - \alpha_* |T - T_c|^q], \quad T_c \simeq 4^\circ\text{C}, \quad q \simeq 1.895.$$

This gives

$$\rho(4^\circ\text{C}) > \rho(3^\circ\text{C}) > \rho(0^\circ\text{C}).$$

Hence, in a 3°C bath, the 0°C interfacial water is lighter than the bulk water and rises along the upward $+x$ direction. In the freshwater-convection simulation, the cold interfacial plume indeed moves upward[10] while gravity points downward along $-x$.

Matched no-flow and freshwater-convection simulations. To isolate the effect of freshwater convection, we run a matched pair of simulations at

$$Ra = 10^6, \quad Pr = 1, \quad T_{\max} = 2.$$

Both simulations use the same grid, timestep, phase-field parameters, active-core temperature, initial ice radius, and final time. The only difference is that buoyancy is disabled in the no-flow case and enabled in the freshwater case.

The matched comparison is shown in Fig. 7. Both simulations show clear net freezing growth. The mean radius grows from

$$S(0) \simeq 0.119991$$

to

$$S(T_{\max}) \simeq 0.135096.$$

The radial growth is

$$\Delta S_{\text{no-flow}} \simeq 1.51047 \times 10^{-2}, \quad \Delta S_{\text{fw}} \simeq 1.51047 \times 10^{-2}.$$

Therefore

$$\frac{\Delta S_{\text{fw}}}{\Delta S_{\text{no-flow}}} \simeq 1.$$

Thus, for the strong-core case, freshwater convection does not appreciably modify the mean freezing rate.

Quantity	No-flow	Freshwater convection
Base grid	$128 \times 128 \times 1$	$128 \times 128 \times 1$
Refined grid	$256 \times 256 \times 1$	$256 \times 256 \times 1$
Rayleigh number	$Ra = 10^6$	$Ra = 10^6$
Prandtl number	$Pr = 1$	$Pr = 1$
Time step	$\Delta t = 2.5 \times 10^{-5}$	$\Delta t = 2.5 \times 10^{-5}$
Final time	$T_{\max} = 2$	$T_{\max} = 2$
Thermal buoyancy	off	on
Freshwater anomaly	off	on
Gravity axis	irrelevant	$gAxis = 1$, gravity along $-x$
Controlled quadrupole forcing	off	off

Table 4: Matched $Ra = 10^6$ simulations used to compare no-flow freezing with freshwater buoyancy-driven convection.

Convection and morphology diagnostics. Although the mean growth is unchanged, the freshwater simulation does develop a nonzero velocity field. The maximum speed remains small,

$$u_{\max} \simeq 2.98 \times 10^{-5}.$$

This indicates that buoyancy-driven flow is present but weak. The angular heat-flux modes and the baseline-corrected interface modes are shown in Fig. 8. At the final time,

$$\xi_2 \simeq 7.83 \times 10^{-6}, \quad \xi_4 \simeq -1.04 \times 10^{-2}.$$

The fourfold heat-flux contribution is also present in the no-flow baseline and is therefore interpreted mainly as a Cartesian-grid/interface-extraction artifact. After subtracting the no-flow baseline at matched radius, the interface modes are

$$M_1 \simeq 1.62 \times 10^{-6}, \quad M_2 \simeq 4.98 \times 10^{-7}, \quad M_4 \simeq 3.31 \times 10^{-7}.$$

These values are far below the visible-deformation threshold. The freezing front therefore remains approximately circular.

The velocity field is visualized in Fig. 9 and Fig. 10. The background shows the physical temperature field, while the black contour marks the ice–water interface $\phi = 1/2$. The streamlines show that freshwater motion is present, but the flow remains weak. The cold interfacial water rises along $+x$, consistent with the freshwater density anomaly.

Pure-conduction wall-forced anisotropy. As a controlled alternative to the freshwater-convection simulations, we also test whether anisotropic interface deformation can be generated by pure conduction alone. Here buoyancy and fluid advection are switched off, and the anisotropy is imposed directly through external thermal forcing.

We consider three matched phase-field simulations. The first is a baseline pure-conduction run with no fixed-temperature wall forcing. The second fixes the two non-periodic AFiD x -walls, $x = 0$ and $x = L_x$, to a warmer temperature $T = 5^\circ\text{C}$. The third keeps the same warm x -walls and additionally adds thermal reservoir strips near the periodic y -boundaries, held at $T = 3^\circ\text{C}$. Since the y -direction is periodic in the standard AFiD geometry, this last case is not a true Dirichlet y -wall boundary condition, but a box-like thermal-reservoir proxy. (See Table 5 for complete set-up)

The target geometry was an initially circular ice seed with radius $r = 0.2$, the common temperatures are

$$T_{\text{core}} = -20^\circ\text{C}, \quad T_\phi = 0^\circ\text{C}, \quad T_\infty = 3^\circ\text{C}.$$

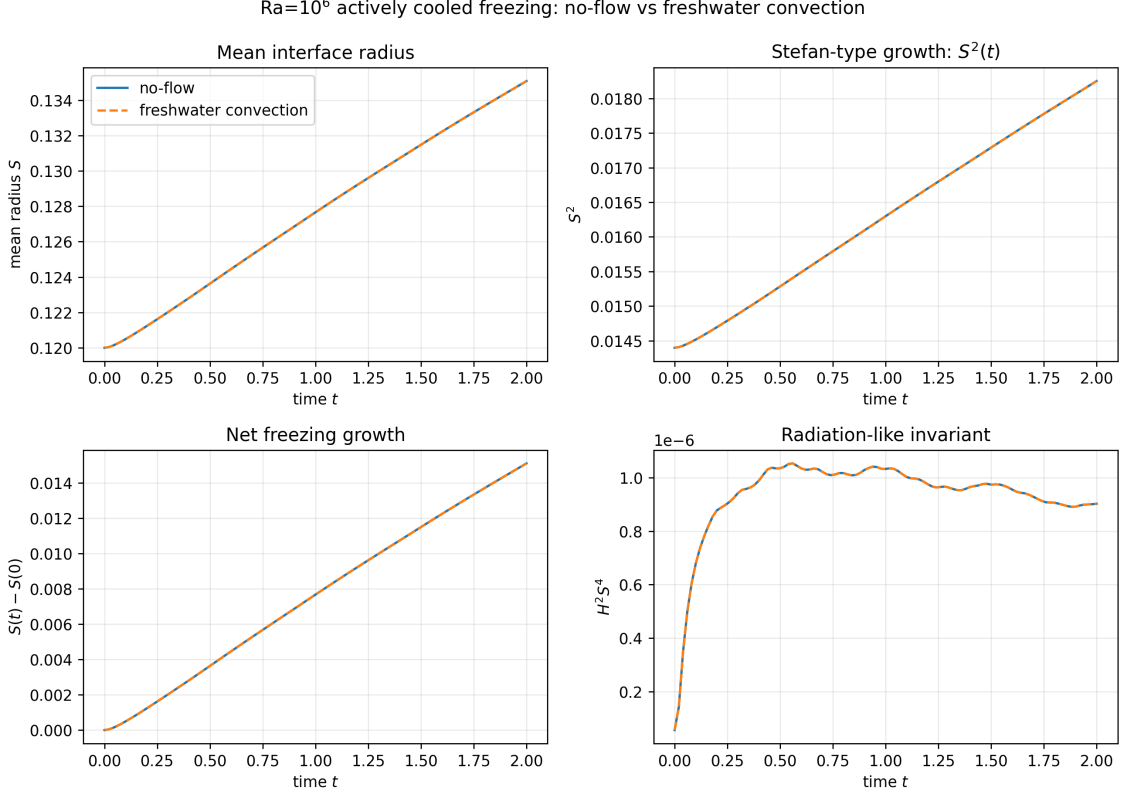


Figure 7: Matched AFiD-MuRPhFi simulations at $Ra = 10^6$ comparing no-flow actively cooled freezing with freshwater-convection freezing. The mean radius $S(t)$, the radius-squared growth $S^2(t)$, the net growth $\Delta S(t)$, and the radiation-like invariant $H^2 S^4$ are nearly identical in the two simulations. Thus, for the strong-core case, freshwater natural convection does not appreciably modify the mean freezing rate.

The nondimensional temperature is

$$\Theta = \frac{T - T_{\text{core}}}{T_{\infty} - T_{\text{core}}},$$

so that

$$\Theta_{\phi} = \frac{20}{23}, \quad \Theta_{x,\text{wall}} = \frac{25}{23}, \quad \Theta_{y,\text{res}} = 1.$$

To isolate the wall-forced response, the baseline mode amplitudes are subtracted,

$$M_n^{\text{corr}} = M_n^{\text{forced}} - M_n^{\text{base}}.$$

A direct axis audit is also performed using the extracted $\phi = 1/2$ contour,

$$R_x = \frac{x_{\text{max}} - x_{\text{min}}}{2}, \quad R_y = \frac{y_{\text{max}} - y_{\text{min}}}{2}.$$

Thus $R_x < R_y$ means that the interface is flattened in the wall-normal x -direction and elongated along the periodic y -direction. The deformation is shown in Fig 11.

At $T_{\text{max}} = 2$, the x -wall case gives

$$M_{2,\text{xwall}}^{\text{corr}} \simeq 7.07 \times 10^{-3}, \quad M_{4,\text{xwall}}^{\text{corr}} \simeq 1.02 \times 10^{-3}.$$

The xy -wall proxy case gives

$$M_{2,\text{xywall}}^{\text{corr}} \simeq 6.09 \times 10^{-3}, \quad M_{4,\text{xywall}}^{\text{corr}} \simeq 1.84 \times 10^{-3}.$$

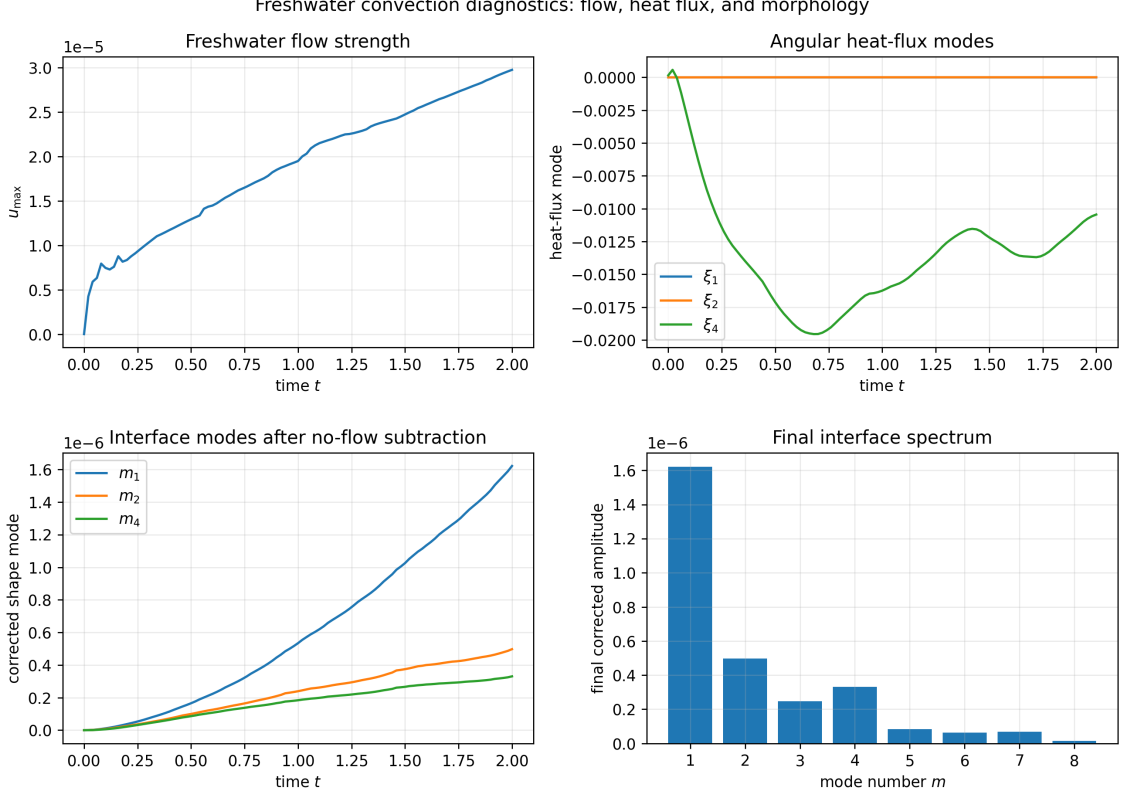


Figure 8: Freshwater-convection diagnostics for the matched $Ra = 10^6$ run. The velocity field is nonzero, confirming the presence of buoyancy-driven flow. However, the baseline-corrected interface modes remain very small. The interface therefore stays approximately circular even though weak freshwater convection is present.

Thus, in both wall-forced simulations, the dominant response is the $n = 2$ elliptic mode rather than the residual fourfold mode (See Fig 12). Adding the y -reservoir strips slightly reduces the quadrupolar amplitude, but it does not change the deformation direction.

The mean freezing growth is also slightly reduced by the external thermal forcing:

$$\Delta S_{\text{base}} \simeq 1.291 \times 10^{-1}, \quad \Delta S_{\text{xwall}} \simeq 1.271 \times 10^{-1}, \quad \Delta S_{\text{xywall}} \simeq 1.270 \times 10^{-1}.$$

For the xy -wall proxy case, the direct axis audit gives

$$(R_x - R_y)_{\text{corr}} \simeq -3.87 \times 10^{-3} < 0,$$

confirming that the warm x -wall forcing produces an x -flattened and y -elongated interface.

These simulations show that a measurable anisotropic freezing response can be generated by pure conduction alone. Compared with the freshwater natural-convection case, the wall-forced setup provides a cleaner and more controllable mechanism for producing a leading $n = 2$ deformation of the freezing front.

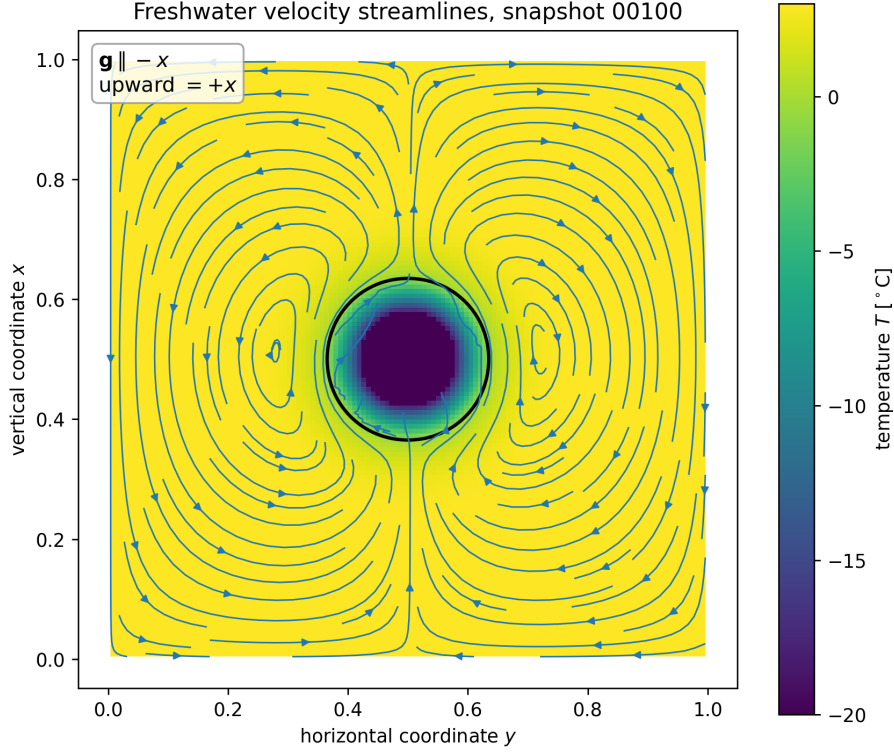


Figure 9: Velocity streamlines in the freshwater-convection simulation at $Ra = 10^6$. The background shows the physical temperature field, and the black contour marks the ice–water interface $\phi = 1/2$. The vertical direction is the AFiD x -direction, with gravity pointing downward along $-x$. The cold interfacial freshwater rises along $+x$, as expected from the freshwater density anomaly below 4°C . The velocity field remains weak, and the interface stays nearly circular.

Quantity	Baseline	x -wall	xy -wall proxy
Core radius	0.195	0.195	0.195
Initial ice radius	0.200	0.200	0.200
Core temperature	-20°C	same	same
Interface temperature	0°C	same	same
Initial liquid temperature	3°C	same	same
x -wall condition	no-flux	fixed 5°C	fixed 5°C
y -side condition	periodic	periodic	3°C reservoir strips
Rayleigh number	$Ra = 10^4$ nominal	$Ra = 10^4$ nominal	$Ra = 10^4$ nominal
Prandtl number	$Pr = 1$	$Pr = 1$	$Pr = 1$
Thermal buoyancy	off	off	off
Freshwater anomaly	off	off	off
Phase field	on	on	on
Final time	$T_{\max} = 2.0$	$T_{\max} = 2.0$	$T_{\max} = 2.0$

Table 5: Matched pure-conduction simulations used to test wall-forced anisotropy. The Rayleigh number is retained only as a nominal nondimensional input; buoyancy is disabled in all three runs.

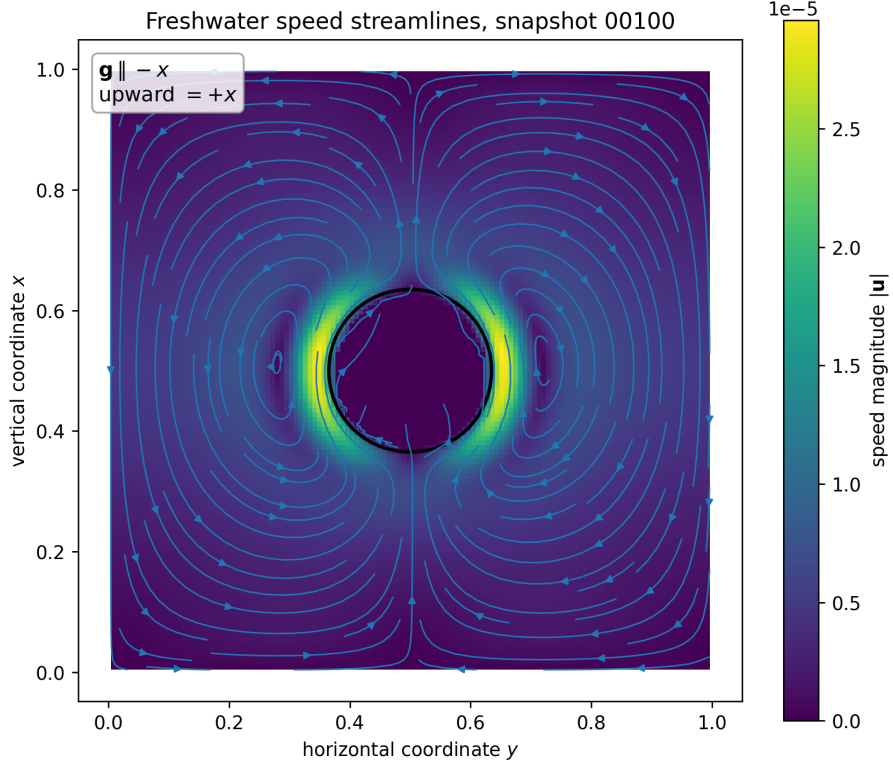
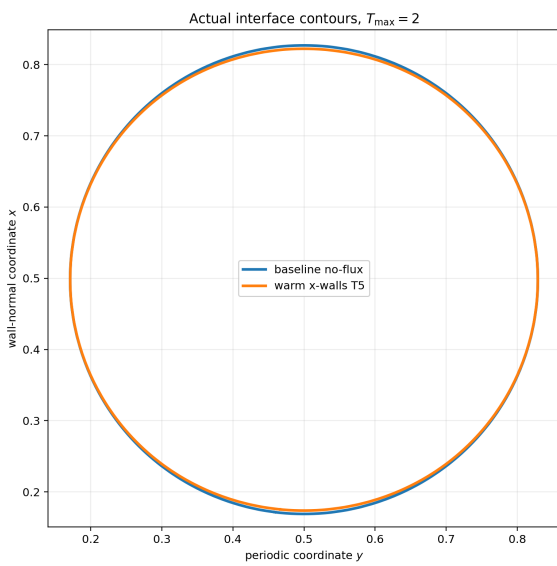
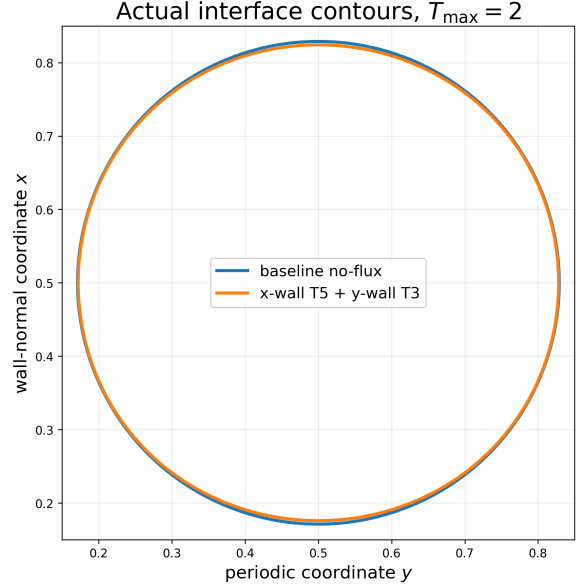


Figure 10: Speed magnitude and streamlines for the freshwater-convection run. Although the velocity field is nonzero, its magnitude remains small. This explains why natural convection does not visibly deform the freezing interface in the strong-core regime.

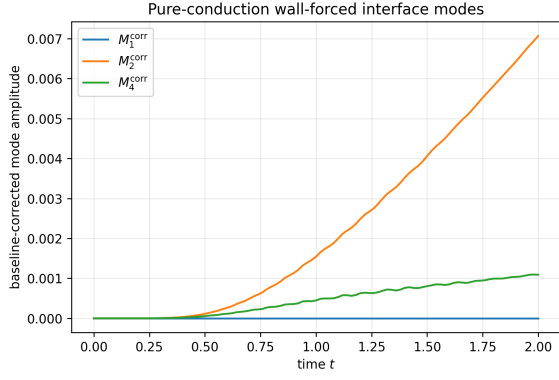


(a) x -wall forcing only.

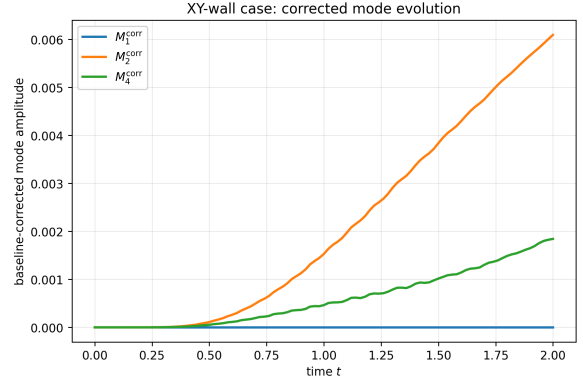


(b) x -wall forcing plus y -reservoir proxy.

Figure 11: Final interface contours at $T_{\max} = 2$ for the two pure-conduction wall-forced simulations. The horizontal axis is y and the vertical axis is x . In both cases the warm x -wall forcing suppresses growth in the wall-normal direction, producing an x -flattened, y -elongated interface.



(a) x -wall forcing only.



(b) x -wall forcing plus y -reservoir proxy.

Figure 12: Time evolution of the corrected interface mode amplitudes. In both wall-forced cases, M_2^{corr} grows and remains larger than M_4^{corr} , showing that the leading wall-induced response is a genuine elliptic $n = 2$ deformation rather than a residual grid artifact.

7 Conclusion

The spherical freezing model gives a direct Stefan analogue of radiation-era cosmology: quasi-static heat conduction leads to $\dot{R} \propto R^{-1}$, hence $R \propto t^{1/2}$ and $H_R^2 \propto R^{-4}$. Introducing an actively cooled isothermal core preserves this scaling while making the setup compatible with a warmer water bath. The reduced anisotropic model shows how water-side transport or imposed conductive anisotropy can act as a quadrupolar source for an anisotropic interface deformation.

The AFiD-MuRPhFi simulations support this picture. The no-flow phase-field runs recover the expected $H^2 S^4$ plateau. Freshwater buoyancy produces the correct upward motion of cold interfacial water, but it does not measurably change the mean freezing rate or the interface shape. By contrast, the pure-conduction wall-forced runs produce a robust $n = 2$ deformation, showing that controlled thermal boundary conditions provide a cleaner route to observable anisotropic freezing.

References

- [1] V. Faraoni, “Analogy between freezing lakes and the cosmic radiation era,” *Phys. Rev. Res.*, vol. 2, p. 013187, Feb 2020.
- [2] M. Vollmer, “The freezing of lakes in winter,” *European Journal of Physics*, vol. 40, p. 035101, apr 2019.
- [3] L. F. Niehof, A. Venkatasubramanian, F. Toschi, and S. Liberati, “Freezing lakes as analogue models of Λ CDM cosmology and beyond,” 2 2026.
- [4] Ö. Akarsu, J. D. Barrow, and N. M. Uzun, “Screening anisotropy via energy-momentum squared gravity: Λ CDM model with hidden anisotropy,” *Phys. Rev. D*, vol. 102, no. 12, p. 124059, 2020.
- [5] G. Ahlers, S. Grossmann, and D. Lohse, “Heat transfer and large scale dynamics in turbulent rayleigh-bénard convection,” *Rev. Mod. Phys.*, vol. 81, pp. 503–537, 2009.
- [6] J. D. Barrow, “Cosmological limits on slightly skew stresses,” *Phys. Rev. D*, vol. 55, pp. 7451–7460, 1997.
- [7] T. Koivisto and D. F. Mota, “Accelerating cosmologies with an anisotropic equation of state,” *Astrophys. J.*, vol. 679, pp. 1–5, 2008.
- [8] T. Koivisto and D. F. Mota, “Anisotropic dark energy: Dynamics of background and perturbations,” *JCAP*, vol. 06, p. 018, 2008.
- [9] AFiD-MuRPhFi Developers, “AFiD-MuRPhFi Documentation.” <https://chowland.github.io/AFiD-MuRPhFi/>, 2025. Accessed: 2026-06-08.
- [10] Z. Wang, L. Jiang, Y. Du, C. Sun, and E. Calzavarini, “Ice front shaping by upward convective current,” *Physical Review Fluids*, vol. 6, p. L091501, 2021.